

Problem

Find the Laplace transform of the following signals:

(a) $f(t) = (2t + 4)u(t)$

(b) $g(t) = (4 + 3e^{-2t})u(t)$

(c) $h(t) = (6 \sin(3t) + 8 \cos(3t))u(t)$

(d) $x(t) = (e^{-2t} \cosh(4t))u(t)$

Step-by-step solution

Step 1 of 4

(a)

The time domain signal is,

$$f(t) = (2t + 4)u(t)$$

Calculate the Laplace transform of the signal, $f(t)$.

$$\begin{aligned} \mathcal{L}\{f(t)\} &= \mathcal{L}\{(2t + 4)u(t)\} \\ F(s) &= 2\mathcal{L}\{tu(t)\} + 4\mathcal{L}\{u(t)\} \end{aligned}$$

Consider the following Laplace transform formulae:

$$\mathcal{L}\{tu(t)\} = \frac{1}{s^2}$$

$$\mathcal{L}\{u(t)\} = \frac{1}{s}$$

Apply the Laplace transform formulae to $F(s)$.

$$\begin{aligned} F(s) &= 2\left(\frac{1}{s^2}\right) + 4\left(\frac{1}{s}\right) \\ &= \frac{2}{s^2} + \frac{4}{s} \end{aligned}$$

Thus, the Laplace transform of the signal, $f(t)$ is $\boxed{\frac{2}{s^2} + \frac{4}{s}}$.

Step 2 of 4

(b)

The time domain signal is,

$$g(t) = (4 + 3e^{-2t})u(t)$$

Calculate the Laplace transform of the signal, $g(t)$.

$$\mathcal{L}\{g(t)\} = \mathcal{L}\{(4 + 3e^{-2t})u(t)\}$$

$$G(s) = 4\mathcal{L}\{u(t)\} + 3\mathcal{L}\{e^{-2t}u(t)\}$$

Consider the following Laplace transform formulae:

$$\mathcal{L}\{e^{-at}u(t)\} = \frac{1}{s+a}$$

$$\mathcal{L}\{u(t)\} = \frac{1}{s}$$

Apply the Laplace transform formulae to $G(s)$.

$$\begin{aligned} G(s) &= 4\left(\frac{1}{s}\right) + 3\left(\frac{1}{s+2}\right) \\ &= \frac{4}{s} + \frac{3}{s+2} \end{aligned}$$

Thus, the Laplace transform of the signal, $g(t)$ is $\boxed{\frac{4}{s} + \frac{3}{s+2}}$.

Step 3 of 4

(c)

The time domain signal is,

$$h(t) = (6\sin(3t) + 8\cos(3t))u(t)$$

Calculate the Laplace transform of the signal, $h(t)$.

$$\mathcal{L}\{h(t)\} = \mathcal{L}\{(6\sin(3t) + 8\cos(3t))u(t)\}$$

$$H(s) = 6\mathcal{L}\{\sin(3t)u(t)\} + 8\mathcal{L}\{\cos(3t)u(t)\}$$

Consider the following Laplace transform formulae:

$$\mathcal{L}\{\sin(at)u(t)\} = \frac{a}{s^2 + a^2}$$

$$\mathcal{L}\{\cos(at)u(t)\} = \frac{s}{s^2 + a^2}$$

Apply the Laplace transform formulae to $H(s)$.

$$\begin{aligned} H(s) &= 6\left(\frac{3}{s^2 + 3^2}\right) + 8\left(\frac{s}{s^2 + 3^2}\right) \\ &= \frac{18}{s^2 + 9} + \frac{8s}{s^2 + 9} \\ &= \frac{8s + 18}{s^2 + 9} \end{aligned}$$

Thus, the Laplace transform of the signal, $h(t)$ is $\boxed{\frac{8s + 18}{s^2 + 9}}$.

Step 4 of 4

(d)

The time domain signal is,

$$x(t) = (e^{-2t} \cosh(4t))u(t)$$

Calculate the Laplace transform of the signal, $x(t)$.

$$\mathcal{L}\{h(t)\} = \mathcal{L}\{(e^{-2t} \cosh(4t))u(t)\}$$

$$H(s) = \mathcal{L}\{(e^{-2t} \cosh(4t))u(t)\}$$

Consider the following Laplace transform formulae:

$$\mathcal{L}\{\cosh(at)u(t)\} = \frac{s}{s^2 - a^2}$$

$$\mathcal{L}\{f(t)e^{-bt}\} = F(s+b)$$

$$\mathcal{L}\{e^{-bt} \cosh(at)u(t)\} = \frac{s+b}{(s+b)^2 - a^2}$$

Apply the Laplace transform formulae to $X(s)$.

$$\begin{aligned} X(s) &= \frac{s+2}{(s+2)^2 - 4^2} \\ &= \frac{s+2}{s^2 + 4s + 4 - 16} \\ &= \frac{s+2}{s^2 + 4s - 12} \end{aligned}$$

Thus, the Laplace transform of the signal, $x(t)$ is $\boxed{\frac{s+2}{s^2 + 4s - 12}}$.